

Response to Text (20%)

Title: Cybersecurity threats are increasing rapidly

Problem statement: With the rapid evolution of cyber threats, both individuals and organisations face increasing challenges in cybersecurity. The growing sophistication of cyberattacks makes them harder to identify and prevent. Therefore, it is crucial to educate the public about cybersecurity practices.

COLUMN 1 Source text	COLUMN 2 Main ideas from the text (extracted from source text)	COLUMN 3 Summarise/Paraphrase	COLUMN 4 Synthesis (combine your own point and another group member's point)
NAME (MATRIC NO): GOE JIE YING (A23CS0224) Text 1 Title: The Importance of Cybersecurity Education in School Source reference: Rahman, N. A. A., Sairi, I. H., Zizi, N. A. M., & Khalid, F. (2020). The importance of cybersecurity education in school. International Journal of Information and Education Technology, 10(5), 378-382.	Statistical reports released by the Cyber Crime and Multimedia PDRM Investigation Division, cases of cyber-love scams or better known as the African Scam are in a state of concern. The number of internet fraud cases in Malaysia increased in 2013, when 1095 cases were reported, compared to only 814 cases in 2012. Additionally, an incident of an 18-year-old Malaysian boy who was arrested for committing an offense under the 1987 Copyright Act for uploading and downloading local music and international films without the owner's consent. The uploaded films included Gravity, Pacific Rim, 47 Ronin, The Hangover 3, We Are the Millers, The Hobbit: The Desolation of Smaug, Ride Along and The Wolverine. Furthermore, according	As cybersecurity threats continue to increase rapidly, cybersecurity education has become a crucial step. If cybersecurity education is neglected, it can lead to many negative effects, especially for teenagers who lack mature thinking, and may even harm a country's economy. According to PDRM Investigation Division, Malaysia saw an increase in fraud cases from 814 in 2012 to 1,095 in 2013. Among them, an 18 years old Malaysian boy was using local music and international films without the owner's permission and caused him to be arrested under the 1987 Copyright Act. Besides, in 2025,	Lam Yoke Yu: Since it's nearly impossible to completely eliminate cybersecurity threats from our lives, we must learn to protect ourselves and stay aware of potential risks. As the saying goes, forewarned is forearmed. Tan Yi Ya: We are currently living in the technology era where the usage of network, data and social media must be utilized to achieve almost everything in life. Therefore, instead of refusing to use them, we must be educated about cybersecurity, and make sure we can protect ourselves online. By knowing the typical actions of traps we could fall into,

	to M. Marimuthu, fraudulent purchase of goods online increased in Malaysia in 2015, with a loss of more than RM4.9 million involving the automobile, housing and tourism sectors. Cybersecurity education is also needed to control addiction to computer games. This addiction certainly has a negative impact. Teenagers spend a lot of time on computers and socialise through their gadgets. Over time, an addiction to online games cannot be avoided, and teenagers' precious time is taken up by addiction to their gadgets. This has a very bad impact on teenagers. Nighttime is spent browsing the internet, which will exacerbate the situation and may even cause teenagers to have health problems. These threats and attacks can come in many forms, and users are not always aware that they are being attacked. It is therefore essential to educate and empower users, especially children, on the safe and responsible use of online resources and platforms, to establish a culture of cybersafety.	Malaysia faced a loss of more than RM4.9 million due to fraudulent purchase of goods online. Addiction to computer games and excessive use of social media are also major problems among teenagers. This habit makes them waste their time and can even cause health issues. Nowadays, hackers and cybercriminals have many ways to attack computers. If teenagers spend so much time indulging in online games and social networking, they will be the most vulnerable group. Therefore, cybersecurity education should be taught from an early age to ensure that everyone has the awareness to protect themselves and others before accessing any online resources and platforms. By having enough awareness of using cybersecurity, our family, friends and country can be protected.	we can be more aware and careful to not get scammed. Teh Ru Qian: Cybersecurity education is critical to protecting kids from online threats such as fraud, hacking and overly frequent use of digital platforms. We can lower dangers, safeguard individual and national security and make sure the next generation uses technology properly and safely by combining awareness with ethical digital behavior.
NAME (MATRIC NO):	Creeper Virus (1971)	The evolution of cybersecurity threats began with the invention	Goe Jie Ying:

LAM YOKE YU (A23CS0233) Text 2 Title: Evolution and Impact of Malware: A Comprehensive Analysis from the First Known Malware to Modern-Day Cyber Threats Source reference: Nilupul, S. A. (2024). Evolution and Impact of Malware: A Comprehensive Analysis from the First Known Malware to Modern-Day Cyber Threats. <i>Cyber Security</i>.	One of the earliest computer viruses ever discovered is sometimes listed as the Creeper virus. At BBN Technologies, Bob Thomas created it in the early 1970s. A self-replicating experimental software called Creeper was created to push the boundaries of computer networks at the time. It showcased the potential of a mobile application that could go across networks rather than having a harmful aim. Moving from one DEC PDP-10 computer to another, the Creeper virus operated over the ARPANET, an early form of the internet. It was a rather harmless virus that infected terminals and displayed the phrase "I'M THE CREEPER: CATCH ME IF YOU CAN." The Reaper software, which was intended to remove Creeper, was developed as a result of this attempt. Important early examples of self-replicating programs and what would eventually become antiviral technology are the Creeper and Reaper programs. They contributed to the establishment of the groundwork for more complex conversations and advancements in distributed software management and network security. 1990s: Increased Sophistication	of the Creeper virus in the early 1970s. It was a self-replicating program created for experimental purposes and did not harm the affected network. In response, the Reaper software was developed, marking the creation of the first antivirus software. Later, in the 1990s, with the widespread use of software, more viruses emerged. For example, the Melissa virus infected Microsoft Word documents and spread through email attachments. Over time, malware evolved to serve various purposes, including financial gain, political motives, and strategic objectives. In the past decade, with the advent of artificial intelligence and machine learning, malware has become more adaptive and harder to detect.	As time goes by, people have gone from having no idea about cybersecurity to having enough knowledge to face this issue. As long as education continues in an orderly manner, the consequences of cybersecurity education will be reduced until everyone has awareness. Tan Yi Ya: Tracing back to the start of cybersecurity threats, we are surprised to find that it was not created for any harm. However, as the technology advances over time, people are now using it for fraud, phishing, attacks and data stealing, which could have caused disasters on everyone on the Internet. In order to prevent it, we must first be aware of it, and make sure to not fall easily for suspicious individuals over online platforms or phone calls. Teh Ru Qian: Cybersecurity risks have evolved dramatically over time. They started off as simple testing but are now employed to steal data or money. These attacks are more difficult to prevent now that AI and smart
---	--	---	--

	The 1990s marked a period of rapid growth in the sophistication of malware. During this decade, email became a primary vector for the spread of viruses. Macro viruses, such as the Melissa virus in 1999, exploited vulnerabilities in software like Microsoft Word to infect documents and spread through email attachments. The Melissa virus, named after an exotic dancer, spread rapidly and caused significant disruption by overwhelming email servers. Targeted Attacks (2000s) Malware evolved further to include tools for sabotage and cyber warfare. Notable examples include the Stuxnet worm, which was designed to disrupt Iran's nuclear program. This marked a shift in malware's purpose from financial gain to political and strategic objectives, highlighting its potential as a weapon in cyber conflicts. Current Trends (2024s) Today, malware continues to evolve with the integration of artificial intelligence and machine learning, making it more adaptive and harder to detect. Cybercriminals are increasingly using malware as a service, allowing even less skilled individuals to launch		technology are in place. To stay safe online, we must constantly enhance our security.
--	---	--	---

	attacks. The focus has also shifted towards exploiting vulnerabilities in cloud services and IoT devices.		
NAME (MATRIC NO): Tan Yi Ya (A23CS0187) Text 3 Title: A Review of Phone Scam Activities in Malaysia Source reference: M. F. Mubarak, S. Yahya and A. F. A. Shaazi, "A Review of Phone Scam Activities in Malaysia," <i>2019 IEEE 9th International Conference on System Engineering and Technology (ICSET)</i> , Shah Alam, Malaysia, 2019, pp. 441-446, doi: 10.1109/ICSEngT.2019.8906491.	This situation had happened to a young male executive from Kuala Lumpur. It started when the young executive got a call from police station in Sabah, Malaysia. The policeman which is a fake person said to the young executive that he got drug trafficking case which needs to be settled and couple of his so-called gang have been caught by police. Then, the caller provides an offer to the young executive whether he wanted to settle the case before they bring it to court. Next, the caller transferred the call to another higher rank police officer which is another fake person. The fake higher rank police officer was trying to negotiate with the young executive until he believe and pay some money to them by using internet banking. However, the young executive did not believe whatever they said because he has never been to Sabah recently and never convict any crime related to drug trafficking. Later, he just terminated the call after he realised that everything was fake.	A young male executive from Kuala Lumpur had received a scam call pretending to be a policeman calling from Sabah Police Station. The scammer tried to trick the young man into believing his crime activity had been exposed and he was in trouble, then wanted him to pay through online transfer. Luckily, the young man did not get deceived because he had neither been to Sabah recently nor participated in any crime related activities. Hence, he terminated the scam call. To his surprise, the scammer has his personal info and the phone number calling was indeed from Sabah Police Station. From the incident, it proves that phone call scams are evolving. Scammers will take advantage of data leaking and can even hide themselves as government agencies by encapsulating their phone number through special measures. The pattern of scam	Goe Jie Ying: Awareness is crucial to prevent telecommunications fraud. Don't panic when receiving any call, find a way to confirm the authenticity of the other party's identity first. Only after the authenticity is confirmed to start finding appropriate solutions. Lam Yoke Yu: Recent scams are more localised and mimic trusted entities that become harder to detect. Thus, each of us should remain calm and think rationally when we receive scam calls that threaten us. Teh Ru Qian: Smarter scam calls are becoming more difficult to identify. Scammers could pretend as representatives of the government or police by using stolen personal information and actual phone numbers. They make stories to scare people and ask for money. Before we trust such calls, we must always be alert and be careful.

	What was very surprising to the young executive is the scammer know who he is and the caller identification number that he got was from the right police station but the person who called was not real policeman. This means that the scammer already has personal information of the young male executive. It could be from other sources such as data leakages. So, we must be careful to share our information especially in public or social media. Another question is how the scammer could use caller identification of the police station. We mentioned earlier, it might be the scammer has used some hacking tools to spoof their caller identification number with another number. This is very serious because it could make some people believe that person who called is real. From this scenario we know that scammer will try to play with their victim emotion by making up a story to scare them or put them into complicated situation. Then, scammer will trick the victim with an easy way out from their miserable problem by asking them to pay some amount of money.	calls will be, first making up a story that puts them in a harsh situation, then involves the transaction of money which is said to solve everything.	
--	---	---	--

NAME (MATRIC NO): TEH RU QIAN (A23CS0191) Text 4 Title: Phishing—A Cyber Fraud: The Types, Implications and Governance. Source reference: Ali, M. M., & Mohd Zaharon, N. F. (2022). Phishing—A Cyber Fraud: The Types, Implications and Governance. <i>International Journal of Educational Reform</i>, 33(1), 101-121. https://doi.org/10.1177/10567879221082966	Phishing is a form of cyber fraud. Fraudsters imitate themselves as someone from legitimate institutions to lure individuals into sharing sensitive data such as personal information, passwords and banking and credit card details (Katkuri, 2018). The fraudsters frequently contact their victims via text messages, telephone or email. The retrieved information is subsequently utilised to access vital accounts and cause financial loss and identity theft. For example, a victim receives an email from the bank and is asked to update the personal details and bank account number. Nevertheless, the login pages are bogus, and the hackers take advantage by stealing the victim's information. Spoofing Email, Spear-Phishing and Whaling. One of the phishing attacks is spoofing email. Emails created with a forged sender address are known as spoofing emails. The emails appear to be messages sent to the victims, and the contents trick the victims into opening the email (Gupta et al., 2016). A functioning Simple Mail Transfer Protocol (SMTP) server and mailing software such as Gmail or Outlook make spoofing emails creation	Phishing is a type of cybercrime in which hackers use fake websites or emails to pretend to be reliable sources, such as a bank, government organization or known business. Their objective is to fraud people into providing private information, such as credit card numbers, passwords for websites or identification numbers. Financial loss, identity theft and illegal access to private accounts or networks are just a few of the serious effects that can result from this type of attack. Both people and organizations may be impacted by the harm, which can be challenging to fix. Therefore, employees and the general public must be taught how to spot suspicious communications in order to stop phishing. The risk of falling for phishing scams can be decreased by training sessions, awareness campaigns, and the awareness of sound cybersecurity practices.	Goe Jie Ying: In order to prevent phishing, necessary education is essential. When setting passwords, the public should establish correct concepts, understand the importance of setting complex passwords to protect their passwords from being easily guessed by others. When someone asks for your password, it is best not to tell them. If necessary, confirm the authenticity of the other party's identity first, preferably face to face, to prevent phishing and fraudulent use of the password. Lam Yoke Yu: Phishing attacks today are more contextual and localised. It mimics trusted entities such as government agencies, financial institutions and e-commerce services, making them harder to identify. Tan Yi Ya: I agree that phishing is a serious cyber security threat. I believe the problem can be tackled by raising awareness among the community, such as having awareness campaigns that emphasizes ways to recognize phishing attempts.
---	---	---	---

	easily achievable. The scammer or fraudster can forge the messageheader fields such as the FROM, REPLY-TO and RETURN-PATH addresses immediately after an email message is created. The email with the entered address appears in the recipient's mailbox after being sent (Sanchez & Duan, 2012). No mechanism is provided by the SMTP to address the authentication process. Thus, spoofing email can easily manipulate the users to reveal their personal information by reading or click-ing the email. Companies should take necessary actions to enhance awareness among employees and avoid phishing threats. For example, companies can conduct phishing assess-ments to increase employees' awareness about phishing. Phishing assessment involves sending deceptive emails to employees within the companies to test their security awareness level. These emails imitate phishing emails with social engineering where hyperlinks, open file attachments and access to provide sensi-tive information are attached in these emails. Ikhsan and Ramli (2019) stated that phishing assessment provides knowledge and simulates real cases to		
--	--	--	--

	employees where they can safeguard themselves when facing real phishing emails in the future.		
--	---	--	--

List of references

- Ali, M. M., & Mohd Zaharon, N. F. (2022). Phishing—A Cyber Fraud: The Types, Implications and Governance. *International Journal of Educational Reform*, 33(1), 101-121. <https://doi.org/10.1177/10567879221082966>
- Nilupul, S. A. (2024). Evolution and Impact of Malware: A Comprehensive Analysis from the First Known Malware to Modern-Day Cyber Threats. *Cyber Security*. <http://dx.doi.org/10.13140/RG.2.2.21647.60325>
- M. F. Mubarak, S. Yahya and A. F. A. Shaazi, "A Review of Phone Scam Activities in Malaysia," *2019 IEEE 9th International Conference on System Engineering and Technology (ICSET)*, Shah Alam, Malaysia, 2019, pp. 441-446, doi: 10.1109/ICSEngT.2019.8906491.
- Rahman, N. A. A., Sairi, I. H., Zizi, N. A. M., & Khalid, F. (2020). The importance of cybersecurity education in school. *International Journal of Information and Education Technology*, 10(5), 378-382.